

ASSIGNMENT

SECTION 1.1

1.

```

ls -lah
total 228K
drwx 21 kali kali 4.0K Apr 12 07:17 .
drwx-r-x-x 3 root root 4.0K Dec 2 21:36 ..
-rw-r--r-- 1 kali kali 2.2K Dec 2 21:36 .bash_logout
-rw-r--r-- 1 kali kali 5.5K Dec 2 21:36 .bashrc
-rw-r--r-- 1 kali kali 3.5K Dec 2 21:36 .bashrc.original
drwxrwxr-x 13 kali kali 4.0K Mar 28 03:27 .cache
drwx-r-x-x 15 kali kali 4.0K Mar 28 03:27 .config
drwxr-xr-x 12 kali kali 4.0K Mar 28 04:03 .Desktop
-rw-r--r-- 1 kali kali 35 Feb 7 22:54 .dmrc
drwx-r-x-x 2 kali kali 4.0K Feb 7 22:54 Documents
drwxrwxr-x 3 kali kali 4.0K Feb 12 02:01 .dotnet
drwxr-xr-x 9 kali kali 4.0K Mar 03:18 Downloads
-rw-r--r-- 1 kali kali 12K Dec 2 21:36 .face
lrwxrwxrwx 1 kali kali 5 Dec 2 21:36 .face.icon → .face
-rw-rw-r-- 1 kali kali 86 Feb 10 08:25 .gitconfig
drwx 2 kali kali 4.0K Feb 7 22:54 .git
-rw-r--r-- 1 kali kali 0 Feb 7 22:54 .gitAuthority
drwx-r-x-x 5 kali kali 4.0K Feb 7 22:54 .local
drwx 5 kali kali 4.0K Feb 11 02:40 .mozilla
drwx-r-x-x 2 kali kali 4.0K Feb 7 22:54 .mozilla
drwx-r-x-x 3 kali kali 4.0K Feb 7 22:54 Pictures
drwx 3 kali kali 4.0K Feb 12 01:59 .pki
-rw-r--r-- 1 kali kali 887 Dec 2 21:36 .profile
drwx-r-x-x 2 kali kali 4.0K Feb 7 22:54 Public
drwx-r-x-x 3 kali kali 4.0K Mar 28 03:28 PyCharmMiscProject
-rw-rw-r-- 1 kali kali 213 Feb 8 02:20 sample-files
drwx 3 kali kali 4.0K Feb 26 09:13 .ssh
-rw-rw-r-- 1 kali kali 1.0K Feb 11 02:19 .sshRegistration
-rw-rw-r-- 1 kali kali 1.0K Feb 11 02:19 .ssh_admin_success
drwx-r-x-x 2 kali kali 4.0K Feb 7 22:54 Templates
drwx-r-x-x 2 kali kali 4.0K Feb 7 22:54 Videos
drwxrwxr-x 4 kali kali 4.0K Feb 12 01:59 .vscode
-rw-r--r-- 1 kali kali 1.0K Mar 03:18 .vscode
-rw-r--r-- 1 kali kali 6.0K Apr 12 07:16 .xsession-errors
-rw-r--r-- 1 kali kali 64K Mar 28 03:34 .xsession-errors.old
-rw-r--r-- 1 kali kali 336 Dec 2 21:36 .zprofile
-rw-r--r-- 1 kali kali 1.3K Mar 23 10:34 .zh_history

```

The command `ls -lah` is used to list all files and directories in the current location, including hidden ones, in a detailed format. It shows important information like file permissions, owner name (kali), file size in a human-readable format (KB/MB), and last modified date. Directories such as `.config`, `Desktop`, and `Downloads` appear in blue, making them easy to identify. Overall, this output helps you understand the structure and contents of your home directory.

2.

```

0 # home/kali/.cache/JetBrains/PyCharm2025.3.3/log/threadDumps-Freeze-20260328-032756-PY-253.31033.139-Unsafe.park-Ssec/threadDump-20260328-032756.txt
1 /home/kali/.Downloads/pycharm-2025.3.3/install-linux.tar.gz
2 /home/kali/.Downloads/pycharm-2025.3.3/plugins/atom-plugin/toolbox/build.txt
3 /home/kali/.Downloads/pycharm-2025.3.3/plugins/cwm-plugin/fe-toolbox.txt
4 /home/kali/.Downloads/pycharm-2025.3.3/plugins/failwindcss/server/version.txt
5 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/ansible/ansible_stdlib_packages.txt
6 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/pycharm/teacmy/README.txt
7 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/pycharm/_jb_for_twisted/twisted/plugins/README.txt
8 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/pycharm/django_name_commands_provider/readme.txt
9 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/pycharm/requirements-tests.txt
10 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py39.txt
11 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py312.txt
12 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py311.txt
13 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/win32-py311.txt
14 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/darwin-py314.txt
15 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/darwin-py312.txt
16 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/win32-py310.txt
17 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py310.txt
18 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/darwin-py39.txt
19 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py313.txt
20 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/win32-py310.txt
21 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py314.txt
22 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py312-py310.txt
23 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/linux-py39.txt
24 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/win32-py39.txt
25 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py313.txt
26 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/linux-py310.txt
27 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/darwin-py310.txt
28 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/darwin-py313.txt
29 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/py312.txt
30 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/common.txt
31 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/darwin.txt
32 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/linux.txt
33 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/linux-py312.txt
34 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/linux-py314.txt
35 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stdlib/@tests/stubtest_allowlists/linux-py311.txt
36 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stubs/pdfr2/@tests/stubtest_allowlist.txt
37 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stubs/pdfr2/@tests/stubtest_allowlist.txt
38 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stubs/uwsgi/@tests/stubtest_allowlist.txt
39 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stubs/uwsgi/@tests/stubtest_allowlist.txt
40 /home/kali/.Downloads/pycharm-2025.3.3/plugins/python-ce/helpers/typedshed/stubs/wrong@python3insprins/@tests/stubtest_allowlist.txt

```

The command `find ~ -name "*.txt"` searches for all .txt files starting from the home directory (~) and scans through every subdirectory recursively. The output displays full file paths of all matching files found in the system. Many results are from PyCharm-related folders, indicating that the IDE has created multiple text files internally. This command is useful when you don't know the exact location of a file but know its name or extension.

3.

```
(kali㉿kali)-[~]  
$ nano scan_results.txt  
  
(kali㉿kali)-[~]  
$ tail -n 20 scan_results.txt  
hi !!  
This is Shruti . I am currently interning in Campus Pe.  
/kali㉿kali)-[~]
```

The command `nano scan_results.txt` opens (or creates) a text file in the Nano editor, allowing you to write or edit content. After saving the file, `tail -n 20 scan_results.txt` is used to display the last 20 lines of the file. The output confirms that the text you entered has been successfully saved. This is commonly used to quickly view recent entries in a file without opening the entire file.

4.

```
(kali㉿kali)-[~]  
$ grep -c "OPEN" scan_results.txt  
1  
/kali㉿kali)-[~]
```

The command `grep -c "OPEN" scan_results.txt` searches for the word "OPEN" inside the file and counts how many times it appears. The `-c` option ensures that only the count is displayed instead of the matching lines. The output `1` indicates that the word "OPEN" is present only once in the file. This is especially useful in cybersecurity tasks, such as counting open ports from scan results.

5.

```
(kali㉿kali)-[~]  
$ mkdir -p security/tools/scanners
```

The command `mkdir -p security/tools/scanners` is used to create multiple nested directories in one step. The `-p` option ensures that any missing parent directories are automatically created without errors. Since there is no output, it means the command executed successfully. As a result, a directory structure `security → tools → scanners` is created efficiently.

6.

```
(kali@kali)-[~]
$ nano exploit.sh

(kali@kali)-[~]
$ chmod 776 exploit.sh
```

The command `chmod 776 exploit.sh` is used to change the permissions of the file `exploit.sh`. Here, 7 gives read, write, and execute permissions to the owner and group, while 6 gives read and write permissions to others. This means the script can be executed by the owner and group members but not by others. It is commonly used to make scripts runnable while controlling access.

7.

```
(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$ hostname
kali

(kali@kali)-[~]
$ pwd
/home/kali
```

The command `whoami` displays the current logged-in username, helping you confirm which user is active. `hostname` shows the name of the system or machine you are working on, which is useful in networked environments. `pwd` (print working directory) displays the full path of the current directory you are in. Together, these commands give basic system and user context.

8.

```
$ find . -mtime -7 -type f
./cache/thumbnails/normal/bf39b9cf24017d0d78b17dbb45253114.png
./cache/thumbnails/normal/6a16c972d470bcbee0b3813d36aa1ce1.png
./cache/thumbnails/normal/9bc6e7d2a531730cab9f68651d0d3923.png
./cache/thumbnails/normal/addb21947385d342a1227d04890d3663.png
./cache/thumbnails/normal/632fcd86bdf84aff9c99cb5aa60cca3.png
./cache/thumbnails/normal/9b5a8f54fb64288f41f836108cfa0f0d.png
./cache/thumbnails/normal/d9d03ef38d5313b54dafd131b2ef601c.png
./xsession-errors
./Desktop/assignment_pics/5th.png
./Desktop/assignment_pics/2nd.png
./Desktop/assignment_pics/1st
./Desktop/assignment_pics/3rd.png
./Desktop/assignment_pics/4th.png
./Desktop/assignment_pics/7th.png
./Desktop/assignment_pics/6th.png
./local/share/gvfs-metadata/home-7bca62fd.log
./local/share/gvfs-metadata/home
./local/share/recently-used.xbel
./local/share/Trash/info/assignment_pics.trashinfo
./exploit.sh
./config/xfce4/xfce4-screenshooter
./config/xfce4/desktop/icons.screen0.yaml
./config/xfce4/xfconf/xfce-perchannel-xml/xfce4-panel.xml
./config/Code/User/workspaceStorage/e4d4b45018accb78fa11478e2496df0f/chatEditingSes
./config/Code/User/globalStorage/State.vscdb
./config/Code/DawnGraphiteCache/data_1
./config/Code/DIPS-wal
./config/Code/TransportSecurity
./config/Code/logs/20260328T040016/network-shared.log
./config/Code/logs/20260328T040016/window1/renderer.log
./config/Code/logs/20260328T040016/window1/network.log
./config/Code/logs/20260328T040016/main.log
./config/Code/GPUcache/data_1
./config/Code/Cache/Cache_Data/9f50bddd978eddd_0
./config/Code/Cache/Cache_Data/d121e2a6335b7510_0
./config/Code/Cache/Cache_Data/index-dir/the-real-index
./config/Code/DawnWebGPUcache/data_1
./config/qterminal.org/qterminal.ini
./config/Thunar/accels.scm
```

The command `find . -mtime -7 -type f` searches for files in the current directory (.) that were modified in the last 7 days. The `-mtime -7` option specifies files changed within the past week, and `-type f` ensures only regular files (not directories) are listed. The output shows the paths of all matching files. This is useful for tracking recent changes or activity

9.

```

└─$ grep -r "password" .
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/05D0E0F6DE46A350891269A036F0D4A36EA0CC09: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/1B8027A362789586363EC146E8AD9185E16E3BFC: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/08D9B7EC15CFAP250C45AD4EDE2D9F6A1EB48E15: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/87011AF94CF3D5414DE0F102ACD08F32E0EA3845: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/53C9FC1EC782DAE1340BDE26E8B92E02D010147E: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/9E348FB260837EDE684ABBF6F21280659ADA367: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/D0EA531158D04D4E8F455C588B17623F76E91815: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/48015CF159DFDD4BE02AE62E91D80C83D7589DA09: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/C688ECE3937F493A51CDCE11355EF660EA243FC: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/EF94D58BAC006F8E2B1928E0FCF11D0D25F23C0E: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/0080532ADAAACE225F45DDE718966CE91F1181A5: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/28667DCC281F0D0FA24599D042D83CC6FA5B2F2D: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/BFA73D3CC399B98DDF1FBAACC945C57CF262A785: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/845E356AC5FA708DA3122D6FF01A00E9DCB9AC5: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/D14872C5ECE1BFD6D962ECCCEFA2AB77323CC61B: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/6638496563D6F9201E9B8FDDAE20ED86FCCD984: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/1C0E1DEA6FB77385C96DEC69AAD37FA568CC3CA: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/B60D380714281EF949372CB9135E7F08F7042EC0: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/DFEFA119B8620C0758382D1A6FE5871CCA5C1F13: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/058D770B84A0AEF2712389866E07A562DE662E9: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/8DC48E24C8866BC8C63A7A6456D99D01DC3F22F: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/271EA2A68E04F55FB432620E9577FBA1E4AE90AF: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/DE3292787D121E7FB7FBCA382DA5FD922EB2F2: binary file matches
grep: ./cache/mozilla/firefox/y583wln9.default-esr/cache2/entries/490268B84C4C7DE59701098ECA7BD648F3D307F9: binary file matches

```

The command `grep -r "password"` searches recursively for the word "password" in files within the current directory and its subdirectories. The `-r` flag enables recursive searching through all folders. It outputs the lines where the word is found along with file names. This is often used in security auditing to locate sensitive information.

10.

```

(kali@kali)-[~]
└─$ uname -a
Linux kali 6.16.8+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.16.8-1kali1 (2025-09-24) x86_64 GNU/Linux

```

The command `uname -a` displays detailed system information in a single line. It includes the kernel name, version, system architecture, hostname, and other OS-related details. This helps identify the operating system and its configuration. It is especially useful in troubleshooting and system analysis.